Energy Optimisation of a Crude Distillation Unit

Technical Report on an Aspen HYSYS Study with Python Pinch-Analysis Tools

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1. Summary

This report describes a project on reducing the energy needed to separate crude oil into useful products in a crude distillation unit (CDU). Crude distillation requires a large amount of heat to bring the feed to separation conditions, while the product and circulating streams leaving the columns carry heat that can be recovered. The objective of the work is to use that heat to preheat incoming crude and so reduce the heating duty supplied by external utilities.

The project has two parts. The first is an Aspen HYSYS process template with an accompanying report. The second is a set of Python tools that export stream data from HYSYS in read-only mode, perform pinch analysis, test the sensitivity of utility targets to the minimum temperature approach, and produce preliminary cost and emissions estimates. Results are written as HTML reports and machine-readable JSON files.

One point of scope should be stated at the outset. The repository contains the Aspen template and report together with Python tools that have passed automated testing. The new tools have not yet been validated against a live HYSYS case. All numerical results quoted in this report that come from the Python tools are demonstration results for a synthetic example and should not be read as measurements of the Aspen model.

2. Function of a Crude Distillation Unit

Crude oil is a mixture of many hydrocarbons with different boiling ranges. A CDU separates this mixture into fractions that serve different purposes, as listed in Table 1.

Fraction	Typical purpose
Naphtha	Gasoline blending or petrochemical feedstock
Kerosene	Jet fuel and kerosene production
Diesel	Diesel fuel production
Atmospheric gas oil	Feed for further refinery processing

Fraction	Typical purpose
Atmospheric residue	Feed to vacuum distillation

Table 1. Principal fractions from atmospheric crude distillation.

Distillation is a physical separation. It depends mainly on differences in volatility and is not intended to break large molecules into smaller ones. Inside a column, rising vapour contacts descending liquid, so that lighter components concentrate toward the top and heavier components toward the bottom.

3. Process Flow

The project documentation describes four main stages: preflash, atmospheric distillation, naphtha stabilisation and vacuum distillation. Crude is preheated and conditioned, passes through the preflash and the furnace, and enters the atmospheric column. Naphtha from the column is sent to the stabiliser. The residue is sent to the vacuum column. The stabiliser and the vacuum column therefore treat different products of the atmospheric column; they are not consecutive units through which the whole crude stream passes.

The roles of the principal sections are as follows.

- Preheating recovers heat from hot process streams before the crude reaches the furnace.
- Desalting and conditioning remove salts and water, which reduces corrosion and fouling.
- The preflash removes some lighter material before the main separation.
- The furnace supplies the remaining heat needed to prepare the column feed.
- The atmospheric column separates the crude into boiling-range fractions.
- Side strippers remove light components from side products so that product specifications are met.
- The naphtha stabiliser removes light ends from the naphtha.
- The vacuum column separates heavy material at reduced pressure, which lowers boiling temperatures and limits thermal cracking.

The project README gives a processing basis of 99,000 barrels per day. This figure is the documented design basis and has not been independently checked as part of this report.

4. Motivation for Energy Optimisation

A CDU has two large energy demands that occur together. The incoming crude must be heated, and hot products and circulating streams must be cooled. If hot streams transfer heat to the incoming crude, both the furnace duty and the external cooling duty are reduced. The central principle is that useful process heat should be recovered before additional energy is purchased from utilities.

Potential heat sources include hot product rundown streams and pumparounds. A pumparound withdraws liquid from a column, removes heat from it and returns it to the column. The heat removed can be used to preheat crude and also contributes to the management of heat within the column.

5. Thermodynamic Basis

The project documentation specifies the Peng-Robinson equation of state. A thermodynamic property package estimates quantities such as phase equilibrium and enthalpy, and Peng-Robinson is widely used for hydrocarbon systems in refinery simulation.

Because crude oil contains too many compounds to model individually, it is characterised from assay data: a true boiling point curve, density and light-end composition. Heavier material is represented by pseudo-components, each of which stands for a boiling-range fraction with estimated properties.

The division of work between the two software components should be kept clear. HYSYS calculates process properties and phase behaviour. The Python pinch tool takes stream properties as supplied input and calculates heat-recovery targets from them. It does not replace the HYSYS thermodynamic model.

6. Pinch Analysis

Pinch analysis determines the theoretical minimum heating and cooling requirement for a defined set of process streams and a chosen minimum temperature approach. Each stream is described by a supply temperature, a target temperature and a heat-capacity flow rate, and is classified as hot or cold. A hot stream is one that requires cooling; a cold stream is one that requires heating. The classification follows the process requirement and not the absolute temperature of the stream.

The sensible heat duty of a stream is

$$Q = CP \times (T_{target} - T_{supply})$$

where CP = m × Cp is the heat-capacity flow rate, m is the mass flow rate and Cp is the specific heat capacity. For a cooling duty the magnitude of the temperature decrease is used. The units are given in Table 2.

Quantity	Symbol	Unit
Mass flow rate	m	kg/s
Specific heat capacity	Cp	kJ/(kg·K)
Heat-capacity flow rate	CP	kW/K
Heat duty	Q	kW

Table 2. Quantities and units used in the pinch calculation.

6.1 Minimum temperature approach

Heat transfer requires a temperature difference between the hot and cold streams. The minimum temperature approach, ΔTmin, is the smallest difference permitted when heat-recovery targets are set. A smaller ΔTmin generally allows more heat recovery and lower utility demand, but requires more exchanger area and higher capital investment. A larger ΔTmin reduces recovery but makes heat transfer easier. The trade-off is therefore between energy cost and exchanger investment.

The tool compares utility targets across a range of ΔTmin values. It does not calculate an economically optimal exchanger network, and a sensitivity study of this kind should not be described as a complete optimisation.

6.2 Calculation method

The tool uses the temperature-interval heat-cascade method, which proceeds as follows.

1. Shift hot-stream temperatures down by ΔTmin/2.
2. Shift cold-stream temperatures up by ΔTmin/2.
3. Sort the shifted temperatures to define the temperature intervals.
4. Calculate the net heat in each interval as $\Delta H_i = (\Sigma CP_{hot} - \Sigma CP_{cold}) \times \Delta T_i$.
5. Cascade this heat from the highest interval downward.
6. Add enough hot utility to remove the largest heat deficit.
7. Take the corrected residual at the bottom of the cascade as the minimum cold utility.

The outputs are the minimum hot utility, the minimum cold utility, the maximum heat recovery, the pinch temperatures, the grand composite curve and the variation of utility targets with ΔTmin. Two energy balances are used as consistency checks:

$Q_{recovery} = Q_{cold\ demand} - Q_{H,min}$

$Q_{recovery} = Q_{hot\ available} - Q_{C,min}$

Both expressions must give the same value.

6.3 Demonstration result

A synthetic example was used so that the calculation could be verified by hand. At ΔTmin = 20 °C the results are those in Table 3.

Result	Value
Minimum hot utility	29 kW
Minimum cold utility	60 kW
Maximum heat recovery	355 kW
Hot-side pinch temperature	100 °C
Cold-side pinch temperature	80 °C

Table 3. Pinch targets for the synthetic example at ΔTmin = 20 °C.

The hot-side and cold-side pinch temperatures differ by exactly ΔTmin, as expected. These values come from the synthetic example and are not measurements from the supplied Aspen case.

7. HYSYS Data Export

The exporter runs on Windows and requires a licensed installation of Aspen HYSYS and the optional pywin32 package. It connects to an already running HYSYS session through COM, using the call win32com.client.GetActiveObject("HYSYS.Application"). A mapping file identifies the inlet and outlet material streams that represent each sensible heating or cooling segment.

For each stream the exporter reads temperature (°C), mass flow (kg/s) and specific enthalpy (kJ/kg). It then computes an effective heat-capacity flow rate:

$CP_{effective} = m \times (h_{out} - h_{in}) / (T_{out} - T_{in})$

This preserves the enthalpy duty of the segment while treating its heat capacity as constant over the segment.

The exporter checks for missing streams, invalid values, inconsistent heating and cooling directions, duplicate stream pairs, and inlet-outlet mass-flow mismatch above 0.1 %. Metadata recorded with each export include the case title, the export time, the source measurements and their units. The exporter is read-only: it does not change equipment settings or start the solver. Its purpose is to remove manual data transfer and make the analysis reproducible.

The exporter does not establish that the HYSYS case has converged, or that a segment is free of phase change. Both conditions must be confirmed by the user before the data are used.

8. Economic and Environmental Estimates

Pinch analysis yields a target for useful heating duty. The fuel actually consumed also depends on furnace efficiency η:

Fuel savings = (Q_baseline − Q_target) / η

In the synthetic scenario the baseline useful duty is 35 kW, the target is 29 kW and the efficiency is 0.85. The fuel saving is therefore (35 − 29) / 0.85 = 7.06 kW. At 8,000 operating hours per year this corresponds to 56.47 MWh per year, and at an energy price of 40 currency units per MWh the annual saving is approximately 2,258.82 currency units. The percentage reduction in useful duty is (35 − 29) / 35 × 100 = 17.14 %, which assumes the same efficiency before and after the change.

Carbon dioxide savings are calculated from a supplied fuel emission factor. Simple payback is the retrofit investment divided by the positive annual cost saving.

The baseline and the target must refer to the same process boundary. Comparing the duty of a single furnace with a utility target for a different set of streams would give a misleading result.

9. Control Considerations

The original documentation describes furnace outlet-temperature control, column pressure control, liquid-level control and side-product flow control. In furnace temperature control, the controlled variable is the furnace outlet temperature, the manipulated variable is the fuel-gas flow, and the main disturbances are changes in crude flow or properties. The controller adjusts fuel flow to hold the temperature at its set point.

Energy recovery must be assessed together with operability. Changing preheat conditions alters furnace demand and column feed conditions, so a final design would require control and dynamic validation. The Python tools do not tune controllers or verify dynamic stability.

10. Reported 12 % Reduction and Testing

The original project documentation reports an energy reduction of 12 %. The new analysis does not assume this figure. It calculates targets from explicit stream data. Confirming the reported reduction would require comparable baseline and retrofit cases in Aspen.

The earlier README code multiplied the baseline duty by 0.88. That demonstrates an assumed reduction and does not calculate or validate one. The heat-cascade implementation replaces it.

The project has 44 automated tests. They cover hand-calculated pinch targets, energy balances, economic unit conversions, invalid inputs, report generation and a simulated COM export. Compatibility with a live HYSYS installation remains to be validated on a licensed Windows system.

11. Limitations

- Segments are modelled with a constant effective CP and sensible heat only.
- The tools contain no model for exchanger sizing, pressure drop, fouling or forbidden matches.
- Convergence of the simulation case is not verified automatically.
- No validation against live HYSYS has been performed in the current Linux environment.
- The economics are screening estimates and do not constitute a detailed investment study.

12. Future Work

- Variable-property segmentation of streams that undergo significant property change.
- Estimation of exchanger area and capital cost.
- Inclusion of fouling constraints.
- Validated comparison of baseline and retrofit Aspen cases.

13. Conclusion

The main contribution of the project is a reproducible path from process-stream data to heat-recovery targets and preliminary economic and emissions estimates. It supports engineering decisions while keeping thermodynamic targets separate from improvements that must still be demonstrated in the process simulation itself.